

Set 14: Survey Results

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Date	08 October 2026
Day	Thursday

Aim

To visualize survey results using a stacked bar chart, a radar chart, and a table, and to build an interactive Tableau dashboard that combines all three.

Dataset: Survey Results

Respondent	Question 1	Question 2	Question 3
1	A	B	B
2	B	C	A
3	C	D	A

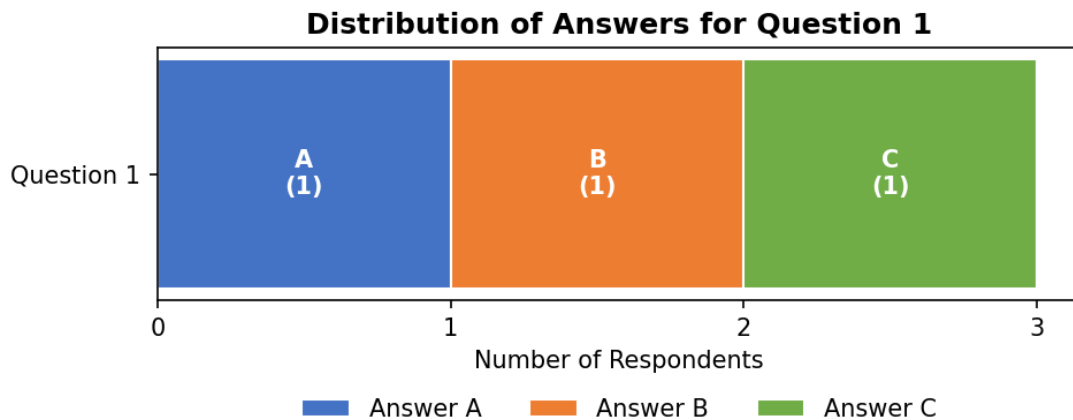
Note: The dataset text was jumbled when copied, so it was read as Respondent 1 = (A, B, B), Respondent 2 = (B, C, A), Respondent 3 = (C, D, A) for Questions 1, 2 and 3. Please check this against your question paper.

PROGRAM 1: Stacked Bar Chart for Question 1

Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

```
# Dataset (assumed reading of the survey table)
survey <- data.frame(
  Respondent = c(1, 2, 3),
  Question1  = c("A", "B", "C"),
  Question2  = c("B", "C", "D"),
  Question3  = c("B", "A", "A")
)

# Stacked bar chart for Question 1
library(ggplot2)
ggplot(survey, aes(x = "Question 1", fill = Question1)) +
  geom_bar(position = "stack") +
  labs(title = "Distribution of Answers for Question 1",
       x = "Question", y = "Number of Respondents",
       fill = "Answer") +
  theme_minimal()
```



Output 1: Stacked Bar Chart of Question 1 Answers

Tableau steps: create a calculated field Number of Records (or use the built-in one) and drag it to Columns. Drag Question 1 to Color and Label, then set the mark type to Bar. Add the title "Distribution of Answers for Question 1" and label the axis "Number of Respondents".

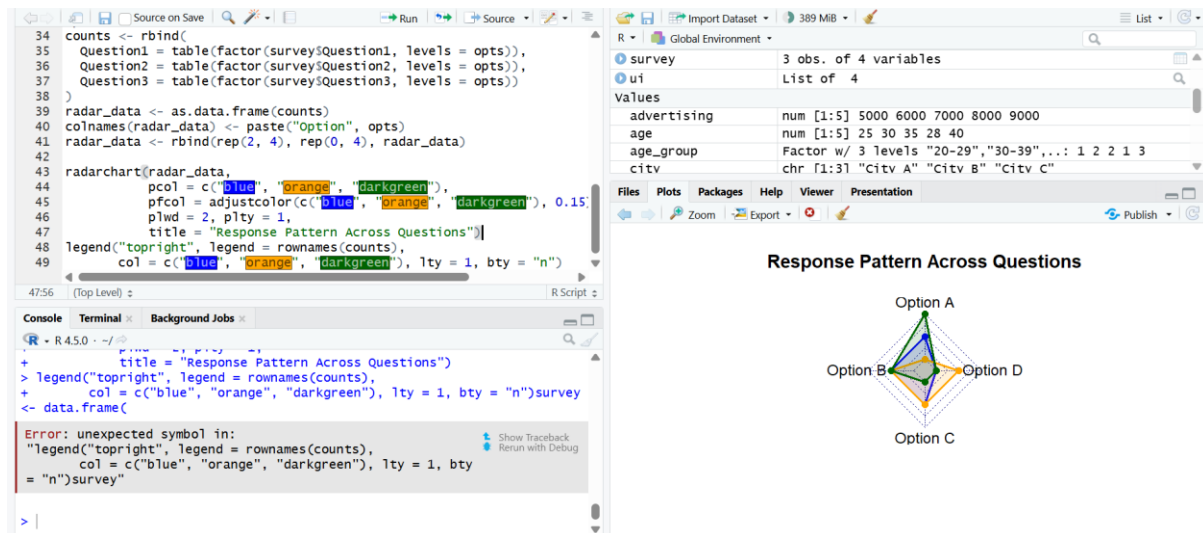
Observation: Each of the answers A, B and C was chosen by exactly one respondent, so the bar is divided into three equal segments (33.3% each). Answer D was not chosen for Question 1.

PROGRAM 2: Radar Chart of Responses

Generate a radar chart to visualize the overall pattern of responses across all three questions.

```
# Radar chart (install.packages("fmsb") if required)
library(fmsb)
opts <- c("A", "B", "C", "D")
counts <- rbind(
  Question1 = table(factor(survey$Question1, levels = opts)),
  Question2 = table(factor(survey$Question2, levels = opts)),
  Question3 = table(factor(survey$Question3, levels = opts))
)
radar_data <- as.data.frame(counts)
colnames(radar_data) <- paste("Option", opts)
radar_data <- rbind(rep(2, 4), rep(0, 4), radar_data) # max, min rows

radarchart(radar_data,
  pcol = c("blue", "orange", "darkgreen"),
  pfcol = adjustcolor(c("blue", "orange", "darkgreen"), 0.15),
  plwd = 2, plty = 1,
  title = "Response Pattern Across Questions")
legend("topright", legend = rownames(counts),
  col = c("blue", "orange", "darkgreen"), lty = 1, bty = "n")
```



Output 2: Radar Chart of Response Pattern Across Questions

Tableau steps: Tableau has no built-in radar chart, so it is built from a polygon. Pivot Question 1–3 into one column (Question, Answer), count the responses for each option, and create calculated fields:

```
Angle = (INDEX() - 1) * 2 * PI() / 4 // 4 options: A, B, C, D
X      = SUM([Count]) * COS([Angle])
Y      = SUM([Count]) * SIN([Angle])
```

Place X on Columns and Y on Rows, set the mark type to Polygon (or Line with Path = Option), put Question on Color/Detail, and add the option labels as text at the axis ends.

Observation: The radar shows how often each option was chosen in each question. Question 3 peaks at Option A (2 responses), Question 2 spreads across B, C and D with one response each, and Question 1 spreads across A, B and C. Option D is selected only in Question 2, and Option A is the most common single answer overall (3 of 9 responses).

PROGRAM 3: Table of Survey Responses

Build a table to show the survey response data for each respondent. Label the table elements.

```
# Table of survey responses
survey_table <- data.frame(
  Respondent = survey$Respondent,
  "Question 1" = survey$Question1,
  "Question 2" = survey$Question2,
  "Question 3" = survey$Question3,
  check.names = FALSE
)
print(survey_table)
View(survey_table)
```

The screenshot shows the RStudio interface. The top pane displays a data table with 4 columns: Respondent, Question 1, Question 2, and Question 3. The bottom pane shows the R console with the following code and output:

```

+ "Question 1" = survey$Question1,
+ "Question 2" = survey$Question2,
+ "Question 3" = survey$Question3,
+ check.names = FALSE
+ )
> print(survey_table)
  Respondent Question 1 Question 2 Question 3
1          1         A         B         A
2          2         B         C         A
3          3         C         D         A
> View(survey_table)

```

The summary statistics for the table are as follows:

Column	Summary
Respondent	<dbl> [1, 3] 0% NA
Question 1	<chr> 3 unique 0% NA
Question 2	<chr> 3 unique 0% NA
Question 3	<chr>

Output 3: Survey Response Table

Tableau steps: create a new sheet, drag Respondent to Rows, and drag Question 1, Question 2 and Question 3 to Rows as dimensions (or use Measure Names/Values), then show them as a text table with headers labelled.

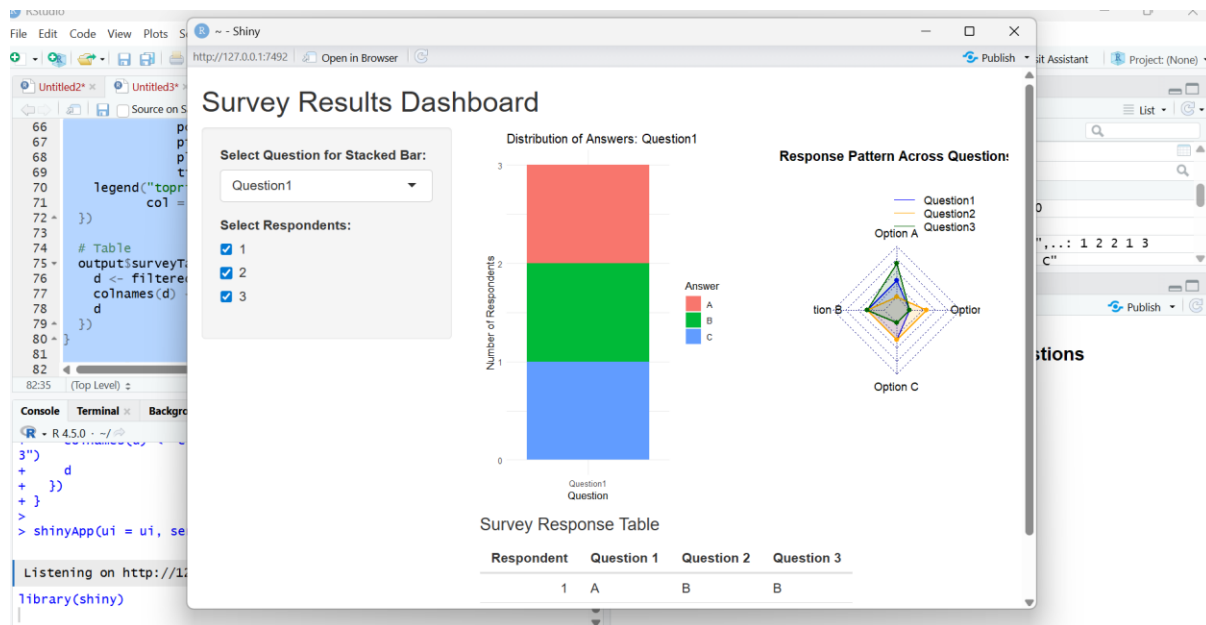
Observation: The table lists the answer each respondent gave for the three questions. Respondents 2 and 3 both chose A for Question 3, and no respondent gave the same answer to all three questions.

PROGRAM 4: Tableau Dashboard

Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

Steps to build the dashboard:

1. Open Tableau and connect to the Excel/CSV file containing Respondent, Question 1, Question 2 and Question 3.
2. Create three worksheets: Stacked Bar (Program 1), Radar Chart (Program 2), and Table (Program 3).
3. Click New Dashboard and set the size to Automatic (or Fixed 1000 x 800).
4. Drag the Stacked Bar to the top left, the Radar Chart to the top right, and the Table across the bottom.
5. For each sheet, click the dropdown arrow and select Use as Filter, so clicking an answer or respondent filters the other views.
6. Add a Respondent quick filter (Show Filter) and apply it to all worksheets.
7. Add the dashboard title "Survey Results Dashboard" and tidy the legends and layout.
8. Save the workbook and use Presentation Mode to test the interactions.



Output 4: Interactive Survey Results Dashboard (Tableau)

Observation: The dashboard combines the stacked bar chart, radar chart, and table in one view. Selecting a respondent or an answer highlights the matching data in the other views, which makes the survey responses easy to explore.

Procedure

1. Create the survey data frame with Respondent and Question 1–3 in R.
2. Use ggplot2 with geom_bar() to create the stacked bar chart for Question 1.
3. Count the answers for each option and use radarchart() from fmsb to create the radar chart.
4. Create a data frame/table to display the responses of each respondent.
5. In Tableau, build the three worksheets, combine them in a dashboard, and enable filters for interactivity.

Result

The survey results were successfully visualized using a stacked bar chart, a radar chart, and a table. An interactive Tableau dashboard was created by combining all three views for exploring the survey responses.